**Table 11: Functions and dietary sources of ultra-trace minerals**

| Mineral | Essential for | Functions | Good dietary sources |
|---|---|---|---|
| Arsenic | Animals possibly humans | Aids normal growth and use of iron; needed for conversion of methionine to cysteine. | Fruits, vegetables, fish, shellfish, grains |
| Boron | Plants Possibly animals Possibly humans | Affects use of calcium, magnesium and copper and protein metabolism; involved in composition of kidney and brain; may be needed for membrane function; may help prevent loss of calcium from bone in postmenopausal women. | Noncitrus fruits, leafy greens, nuts, legumes, cider, wine, beer. |
| Cobalt | Animals Possibly humans | Is part of vitamin $B_{12}$, which is necessary for production of normal red blood cells. | Liver and red meat. |
| Nickel | Chickens, rats, pigs, and goats possibly humans | Involved in iron absorption, use of calcium, zinc, and vitamin $B_{12}$, and metabolism of genetic material (DNA) | Nuts, legumes, whole grains. |
| Silicon | Animals possibly humans | Probable required for deposition of minerals, especially calcium, in bones; needed for synthesis of collagen and elastin, major proteins in connective tissue. | Whole grains, cereal products, root vegetables (e.g. potatoes, carrots) |
| Tin | Rates | Is necessary for normal growth | Commercial fats |
| Vanadium | Experimental animals | Probably involved in iodine and glucose metabolism and thyroid function | Whole grains and grain products; meat, poultry, and fish (moderate levels) |

## 5.10 Therapeutic Diets

### Modification of nutrients in therapeutic Diets

The general principles of nutrition related to health apply also to the treatment of patients suffering from various diseases. Diet in disease must be planned as part of the complete care of the patient many modifications may have to be made according to the disease and the condition of the patient, but there are certain general principles which may be used for guidance.

### Principle of Therapeutic Diets

1. Carbohydrates are usually well – tolerated and are necessary to maintain the stores of liver glycogen.
2. The tolerance of fats varies in different individuals and this nutrient should not be forced if there is nausea and vomiting.
3. In illness, there is usually an increased demand for proteins, due to wasting and this should be given in easily digestible forms such as milk, egg, chicken and fish.
4. The requirements of calcium and iron must be maintained during illness and it is necessary to check.
5. Fat – soluble vitamins often need to be added as concentrates if a patient has to be on a fat – restricted diet for a long time. The demand for Vitamin C is greatly increased in fevers, and it is especially necessary for the healing of wound after surgery.



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